Solving Exponential and Logarithmic Equations
Answer Key

Algebra 2   Exponential and Logarithmic Functions

Quick Answers

1. 4
2. 2.322
3. 32
4. $\frac{4\log(x)}{\log(3)} + 2$
5. 18
6. 7.33
7. 5
8. $\frac{\log\left(\frac{x^2}{3}\right)}{\log(2)}$
9. $\frac{3}{2}$
10. 3

Curriculum

Level: Algebra 2   Exponential and Logarithmic Functions
HSF-BF.B.5 – *problems 3, 4, 7, 8, 10*
HSF-LE.A.4 – *problems 1, 2, 5, 6, 9*
Difficulty 1–4 of 5 across 10 tagged checks.

1. Yeast culture: solve $3^x = 81$.

Step 1. Both sides can be written as powers of the same base, so no logarithm is needed — matching exponents is exact and faster.
Step 2. $81 = 3^4$, so the equation reads $3^x = 3^4$. An exponential function is one-to-one, so the exponents must be equal.
Step 3. The culture reaches that size after $\boxed{\text{4 hours}}$
Check: $3^4 = 81$, and the model says the size is a multiple of the start, so the answer is a time, not a size.

2. Bacteria: solve $2^x = 5$ to the nearest thousandth.

Step 1. Five is not a power of two, so the bases cannot be matched. Take a logarithm of both sides and use the power law to bring the unknown down.
Step 2. $\log(2^x) = \log 5$ gives $x \log 2 = \log 5$, so $x = \dfrac{\log 5}{\log 2}$.
Step 3. Evaluating, $x = 2.32192\ldots$, so the population reaches five times its start after $\boxed{\text{2.322 days}}$
Common wrong answer: 2.5, from dividing the two numbers instead of their logarithms. $\log 5 \div \log 2$ is not $5 \div 2$; the logarithms must be taken first.

3. Doubling-time model: solve $\log_2 n = 5$.
Step 1. A logarithm is an exponent, so the fastest route is to read the equation as its exponential twin rather than to manipulate it.
Step 2. $\log_2 n = 5$ means $2^5 = n$.
Step 3. $2^5 = 32$, so $\boxed{n = 32}$
Teaching note: say the sentence out loud — "two to the fifth is n". Students who narrate the rewrite stop reaching for a calculator on problems that do not need one.

4. Expand $\log_3(9x^4)$.
Step 1. The argument is a product, and one factor carries a power, so the product law comes first and the power law second.
Step 2. Product law: $\log_3(9x^4) = \log_3 9 + \log_3(x^4)$. Power law on the second term: $\log_3(x^4) = 4\log_3 x$.
Step 3. $\log_3 9 = 2$ because $3^2 = 9$, so the expansion is $\boxed{2 + 4\log_3 x}$
Listen for: $\log_3 9 = 3$. Ask what exponent turns 3 into 9, not what number goes into it.

5. Half-life: solve $200\left(\frac{1}{2}\right)^{t/6} = 25$.
Step 1. Isolate the exponential factor before doing anything with the exponent — the constant in front is not part of the power.
Step 2. Dividing by 200 gives $\left(\frac{1}{2}\right)^{t/6} = \frac{1}{8}$, and $\frac{1}{8} = \left(\frac{1}{2}\right)^3$, so $\dfrac{t}{6} = 3$.
Step 3. Multiplying by 6, the sample falls to that mass after $\boxed{18 \text{ years}}$
Sense check: three half-lives take the sample $200 \to 100 \to 50 \to 25$, which is exactly three periods of 6 years.

6. Account: solve $1500(1.04)^t = 2000$ to the nearest hundredth.
Step 1. Isolate the power, then take logarithms — 1.04 and the ratio of the balances share no convenient common base.
Step 2. $(1.04)^t = \dfrac{2000}{1500}$, so $t\log(1.04) = \log\left(\dfrac{2000}{1500}\right)$ and $t = \dfrac{\log(2000/1500)}{\log(1.04)}$.
Step 3. Evaluating gives $t = 7.33495\ldots$, so the balance reaches that level after $\boxed{7.33 \text{ years}}$
Common wrong answer: 1.33, the ratio of the balances. That ratio is the growth *factor* needed, not the time; the logarithm converts a factor into a number of periods.

7. Solve $\log x + \log(x - 3) = 1$.
Step 1. Two logarithms with the same base and a plus sign between them condense by the product law, which leaves one logarithm to undo.
Step 2. $\log(x(x-3)) = 1$, and in exponential form that is $x(x-3) = 10^1$. Expanding, $x^2 - 3x - 10 = 0$, which factors as $(x-5)(x+2) = 0$, so the candidates are $x = 5$ and $x = -2$.
Step 3. A logarithm needs a positive argument. At $x = -2$ both $\log x$ and $\log(x-3)$ are undefined, so that candidate is extraneous — introduced by condensing, not by the original equation. The solution is $\boxed{x = 5}$
Check: $\log 5 + \log 2 = \log 10 = 1$, which confirms both the condensing and the rejected root.

8. Condense $2\log_2 x - \log_2 3$.

Step 1. A coefficient in front of a logarithm is a power inside it, so clear the coefficient before touching the subtraction.

Step 2. Power law: $2\log_2 x = \log_2(x^2)$. The expression is now $\log_2(x^2) - \log_2 3$, and a difference of logarithms with the same base is the logarithm of a quotient.

Step 3. Quotient law gives $\boxed{\log_2\left(\frac{x^2}{3}\right)}$

Why it matters: once the whole side is a single logarithm, the equation can be rewritten in exponential form in one move. Condensing is the enabling step, not decoration.

9. Error analysis: $4^{x+1} = 32$.

Step 1. Dividing 32 by 4 treats the equation as if the base and the power were multiplied together. They are not: 4^{x+1} means repeated multiplication of 4, so dividing by 4 removes one factor, it does not remove the exponent.

Step 2. Both 4 and 32 are powers of 2: $4 = 2^2$ and $32 = 2^5$. The equation becomes $(2^2)^{x+1} = 2^5$, that is $2^{2x+2} = 2^5$.

Step 3. Matching exponents gives $2x + 2 = 5$, so $\boxed{x = \frac{3}{2}}$

Check: $4^{5/2} = \left(\sqrt{4}\right)^5 = 32$, and Dev's answer would give 4^8, which is far larger — substituting once would have caught it.

10. Solve $2\log x = \log(x+6)$.

Step 1. Both sides are single logarithms with the same base once the coefficient on the left is absorbed, and equal logarithms force equal arguments.

Step 2. $2\log x = \log(x^2)$, so $\log(x^2) = \log(x+6)$ gives $x^2 = x + 6$. Then $x^2 - x - 6 = 0$ factors as $(x-3)(x+2) = 0$, with candidates $x = 3$ and $x = -2$.

Step 3. The original equation contains $\log x$, which is undefined for a negative argument, so $x = -2$ is rejected and the solution is $\boxed{x = 3}$

Worth naming: $x = -2$ does satisfy $x^2 = x + 6$. The extraneous root appears because squaring inside the logarithm widened the domain — the rejection is a domain fact, not an arithmetic slip.